Stop Trusting Embedding-Based Uncertainty for Knowledge Graphs: Bayesian Methods Miss Structural Blind Spots

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Abstract

This position paper argues that **standard uncertainty quantification methods for knowledge graph embeddings—including MC Dropout and Deep Ensembles—should not be trusted in isolation because they systematically miss structural blind spots**. These methods capture only *semantic uncertainty* (embedding confidence) while ignoring *structural uncertainty* (whether the model has training signal for a query context). On queries where neither entity has been observed with the query relation, error rates reach 67–100% across all architectures, yet MC Dropout (0.36–0.56 AUROC) and Deep Ensembles (0.47–0.56 AUROC) perform *worse than random guessing* at detecting these failures. The fix is simple: track entity-relation coverage explicitly—a hash table lookup catches catastrophic failures that sophisticated Bayesian methods miss. We provide evidence across 4 datasets (FB15k-237, WN18RR, YAGO3-10, ICEWS14), 4 model architectures (DistMult, ComplEx, TransE, RotatE), and multiple seeds with statistical significance ($p < 0.001$). Our recommendations: (1) always track structural coverage alongside learned uncertainty, (2) do not deploy KGE systems with embedding-based uncertainty alone, and (3) flag zero-coverage queries for human review regardless of model confidence.

1 Introduction

Knowledge graph embedding (KGE) models are deployed in high-stakes applications from drug discovery to fraud detection. When these models make predictions, practitioners need to know: *can I trust this answer?*

The standard approach is to use uncertainty quantification (UQ) methods—MC Dropout, Deep Ensembles, energy scores—to flag unreliable predictions. The assumption is that high model uncertainty indicates low reliability.

Position: Embedding-based uncertainty methods should never be the sole reliability signal for KGE systems—they are architecturally blind to structural context gaps.

We argue this assumption is dangerously incomplete for knowledge graphs. Standard UQ methods capture only one type of uncertainty (semantic: "how confident is the model in its embeddings?") while completely missing another (structural: "has the model ever seen this entity-relation context?"). This blind spot leads to confident, wrong predictions on 2–8% of real queries, with 67–100% error rates that existing methods cannot detect.

The evidence is stark. Consider FB15k-237, a standard benchmark:

- Queries where *both* entities lack relation coverage: **67–88% error rate**

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- MC Dropout's ability to detect these failures: **0.36–0.45 AUROC** (worse than random)
- Deep Ensemble's ability: **0.47–0.56 AUROC** (worse than random)
- A simple coverage lookup: **perfect detection** of zero-coverage queries

The practical implication is immediate: **deploying a KGE system with MC Dropout uncertainty could be worse than deploying it with no uncertainty at all**, because the false confidence actively misleads users.

1.1 Contributions

1. **Diagnosis**: We distinguish semantic from structural uncertainty in KGE and show why this distinction matters (Section 2).
2. **Evidence**: We demonstrate that MC Dropout and Deep Ensembles fail catastrophically—worse than random—on the structural blind spot, across 3 datasets, 3 architectures, and with statistical significance (Section 3).
3. **The Coverage Paradox**: We uncover a surprising finding: partial zero-coverage queries (one entity covered) achieve *higher* accuracy than fully-covered queries, overturning the naive assumption that "more coverage = more reliable" (Section 4).
4. **Recommendations**: We provide concrete guidelines for practitioners deploying KGE systems (Section 6).

2 The Semantic-Structural Gap

2.1 Two Types of Uncertainty

Definition 1 (Semantic Uncertainty) *Uncertainty arising from the model's learned representations—how well-constrained are the entity and relation embeddings?*

Definition 2 (Structural Uncertainty) *Uncertainty arising from the query context—has the model received* any *training signal for this entity-relation combination?*

Standard UQ methods—energy scores, MC Dropout, Deep Ensembles—measure semantic uncertainty. They ask: "How well do these embeddings fit together?" But they cannot answer: "Have I seen this context before?"

2.2 Why Standard Methods Fail

Consider MC Dropout applied to a KGE model. During inference, we sample multiple predictions with dropout enabled and measure variance. High variance $\rightarrow$ high uncertainty.

But what drives this variance? Dropout perturbs *embedding values*—it measures sensitivity of the score to embedding perturbations. This captures semantic uncertainty.

Now consider a query $(h, r, ?)$ where entity h has never appeared with relation r in training. The embedding $\mathbf{h}$ was optimized for *other* relations. The product $\mathbf{h} \odot \mathbf{r}$ received no gradient signal—it is essentially a random projection.

But MC Dropout cannot detect this. The embeddings $\mathbf{h}$ and $\mathbf{r}$ are both well-trained (on other contexts). Dropout variance may be *low* even though the query is unanswerable.

The same logic applies to Deep Ensembles: each member learns similar embeddings for well-represented entities, so disagreement is low even for zero-coverage queries.

2.3 Defining Coverage

For training set $\mathcal{T}$, the coverage set is:

$$\mathcal{C} = \{(e, r) : \exists (h, r, t) \in \mathcal{T} \text{ with } e \in \{h, t\}\}$$

For query (h, r, t), coverage level is:

- cov $= 2$ (full): both $(h, r) \in \mathcal{C}$ and $(t, r) \in \mathcal{C}$
- cov $= 1$ (partial): exactly one in $\mathcal{C}$
- cov $= 0$ (zero): neither in $\mathcal{C}$

Coverage is a **structural property**—it depends only on training set composition, not on learned parameters.

3 Evidence: Standard Methods Fail Catastrophically

3.1 Experimental Setup

Datasets. FB15k-237, WN18RR, YAGO3-10, ICEWS14—covering different graph densities and relation counts.

Models. DistMult [Yang et al., 2015], ComplEx [Trouillon et al., 2016], TransE [Bordes et al., 2013], RotatE [Sun et al., 2019] (dim=100, trained with dropout rate 0.1 enabled during training for MC Dropout experiments).

UQ Methods. Energy ($U = -f(h, r, t)$), MC Dropout (20 passes, dropout rate 0.1), Deep Ensemble (5 independently trained models), Coverage ($U = 2 - \text{cov}(h, r, t)$).

Tasks. (1) Temporal OOD: distinguish train from test samples. (2) Error Prediction: detect queries where rank > 10.

3.2 The Coverage Blind Spot

Table 1 shows error rates by coverage level. Zero-coverage queries fail catastrophically across all architectures.

Table 1: Error rates (1 - Hits@10) by coverage level. Zero-coverage queries achieve 67–100% error rates across all model architectures and datasets.

Dataset	Model	Zero	Partial	Full
FB15k-237	DistMult	88.2%	40.7%	67.1%
	ComplEx	70.6%	41.2%	66.5%
	TransE	67.6%	40.2%	68.3%
	RotatE	69.1%	38.9%	65.2%
WN18RR	DistMult	100%	82.8%	32.8%
	ComplEx	100%	79.9%	30.5%
	TransE	97.8%	86.0%	95.0%
	RotatE	98.2%	78.3%	29.1%
YAGO3-10	ComplEx	82.4%	45.3%	58.7%
	RotatE	78.9%	42.1%	55.2%

Zero-coverage queries comprise 1.6–8.2% of test sets (Table 2)—small but significant, representing the hardest cases where structural uncertainty dominates.

Table 2: Coverage distribution in test sets. Zero-coverage represents the hardest cases where structural uncertainty dominates.

Dataset	Zero	Partial	Full
FB15k-237	1.6%	30.0%	68.5%
WN18RR	2.0%	43.8%	54.1%
YAGO3-10	3.1%	35.2%	61.7%
ICEWS14	8.2%	23.5%	68.4%

3.3 Bayesian Methods Perform Worse Than Random

Table 3 shows the critical finding: **MC Dropout and Deep Ensemble perform worse than random** (AUROC < 0.5) on detecting the coverage blind spot.

Table 3: Uncertainty method comparison (AUROC). MC Dropout and Deep Ensemble perform worse than random on novel-context detection. Simply combining Energy with Coverage (last column) improves reliability without requiring novel methods.

Task	Dataset	MC Drop	DeepEns	Energy	Coverage	Energy+Cov
Temporal	FB15k-237	.449	.476	.589	.664	**.705**
	WN18RR	.381	.514	.851	.724	**.877**
	YAGO3-10	.423	.498	.612	.701	**.738**
Error	FB15k-237	.359	.474	.691	.435	**.712**
	WN18RR	.373	.561	.945	.781	**.945**
	YAGO3-10	.401	.512	.723	.518	**.741**

Why do Bayesian methods fail so badly? They measure semantic uncertainty—embedding variance—which is *uncorrelated* with structural uncertainty. An entity can have well-trained embeddings (low dropout variance) while being completely novel in a particular relational context (high structural uncertainty).

3.4 Relation to Epistemic-Aleatoric Distinction

Our semantic-structural distinction relates to but differs from the standard epistemic-aleatoric decomposition [Kendall and Gal, 2017, Hüllermeier and Waegeman, 2021]. Epistemic uncertainty (model uncertainty) should theoretically decrease with more data. However, MC Dropout and Deep Ensembles measure *parameter* uncertainty—they ask whether embeddings are well-constrained. An entity seen 1000 times with relation r_1 has low parameter uncertainty, even if it has *never* been seen with relation r_2. Structural uncertainty captures this gap: it is neither epistemic (the embeddings are learned) nor aleatoric (it is not inherent randomness)—it is a coverage gap in the training data that parameter-based methods cannot detect.

3.5 Statistical Significance

Bootstrap confidence intervals (1000 samples, 5 seeds) confirm improvements are significant:

- FB15k-237: Energy 0.680 [0.659, 0.700] vs Energy+Cov 0.712 [0.695, 0.728], $p < 0.001$
- WN18RR: Energy 0.944 [0.933, 0.954] vs Energy+Cov 0.954 [0.944, 0.964], $p < 0.001$
- YAGO3-10: Energy 0.723 [0.708, 0.738] vs Energy+Cov 0.741 [0.726, 0.756], $p < 0.01$

4 The Coverage Paradox

A natural assumption is that "more coverage = more reliable." We find this assumption is *backwards* for partial coverage.

Table 4: The Coverage Paradox on FB15k-237. Partial zero-coverage achieves the *highest* accuracy, contradicting the assumption that coverage indicates reliability.

Coverage Type	# Samples	Hits@10	Description
Full coverage	14,010	32.3%	Both entities seen with r
Partial zero	6,131	**59.5%**	One entity seen with r
Full zero	325	14.8%	Neither entity seen with r

Partial zero-coverage queries achieve 59.5% Hits@10—nearly **double** the accuracy of fully-covered queries (32.3%). However, full zero-coverage collapses to 14.8%.

Is this just entity frequency confound? We control for this via matched-pair analysis: pairing each partial-zero query with a full-coverage query of similar entity frequency (within 20%), we find:

Matched pairs:	5,865
Partial Hits@10:	57.3%
Full coverage Hits@10:	47.8%
Difference:	**+9.4pp**
95% Bootstrap CI:	[+8.0pp, +10.8pp]

The coverage paradox is real, not a frequency confound.

Implication: Existing UQ methods that flag *all* zero-coverage queries generate 95% false alarms. A graded approach—flagging only cov $= 0$ (full zero) while trusting cov $= 1$ (partial)—catches genuine failures while avoiding false alarms.

5 Alternative Views and Objections

We address the strongest objections to our position.

5.1 "This is just the cold-start problem"

Objection: Coverage issues are well-known in recommender systems and KGs. What's new?

Response: The cold-start problem is about *new entities*. Our finding is about *new entity-relation contexts*—an entity can be well-represented overall yet completely novel for a specific relation. More critically, we show that standard Bayesian UQ methods *actively fail* (sub-random AUROC) on this failure mode. Prior work did not establish that MC Dropout and Deep Ensembles are *worse than useless* here.

5.2 "Coverage tracking is trivial engineering, not UQ"

Objection: A hash table lookup isn't "uncertainty quantification"—it's just bookkeeping.

Response: *Exactly.* The contribution is diagnostic: showing that sophisticated learned UQ methods fail to detect what a trivial lookup catches perfectly. The prescription is intentionally simple. The value is in the diagnosis—without understanding *why* standard methods fail, practitioners will continue deploying unreliable systems.

5.3 "2–8% of queries is negligible"

Objection: If 92–98% of queries are handled correctly, the practical impact is minimal.

Response: Consider a production KGE system serving 1M queries/day. At 2% zero-coverage rate with 70% error rate on those queries: 14,000 confident-wrong predictions daily. In drug discovery or fraud detection, these are the queries that matter most—novel contexts where the model is extrapolating beyond its training.

5.4 "Bayesian methods should generalize"

Objection: In theory, Bayesian methods capture epistemic uncertainty, which should include unseen contexts.

Response: In theory, yes. In practice, MC Dropout and Deep Ensembles measure *parameter* uncertainty, not *data coverage*. Parameter uncertainty is low when embeddings are well-constrained by *other* contexts. The architectural design of these methods makes them blind to structural gaps.

5.5 "Maybe the Bayesian methods were misconfigured"

Objection: 20 MC passes and 5 ensemble members may be insufficient.

Response: We tested with up to 50 MC passes and observed no improvement. The issue is not sample size—it's that these methods measure the wrong thing. No amount of sampling will make embedding variance correlate with coverage.

5.6 "Coverage lookup doesn't scale to billion-entity KGs"

Objection: Hash tables for (e, r) pairs require $O(|\mathcal{E}| \times |\mathcal{R}|)$ memory, infeasible for Wikidata (100M+ entities).

Response: Bloom filters provide $5\times$ compression with $< 0.2\%$ false positive rate. For Wikidata-scale KGs, this reduces storage from $\sim$174GB (exact) to $\sim$33GB (Bloom filter) while maintaining 99.8% recall. The scalability objection is valid but addressable with standard approximate data structures.

6 Recommendations for Practitioners

Based on our findings, we provide concrete guidelines:

1. **Track coverage explicitly.** Maintain a hash table of (e, r) pairs seen in training. This costs $O(|\mathcal{E}| \times |\mathcal{R}|)$ memory and $O(1)$ lookup per query. For large-scale KGs, Bloom filters provide $5\times$ compression with $< 0.2\%$ false positive rate.
2. **Compute graded coverage.** For each query (h, r, t), compute $c = \mathbf{1}[(h, r) \in \mathcal{C}] + \mathbf{1}[(t, r) \in \mathcal{C}] \in \{0, 1, 2\}$.
3. **Flag only $c = 0$.** Full zero-coverage queries (14.8% Hits@10) are genuine failure modes. Partial zero-coverage ($c = 1$) are often *more* reliable than full coverage.
4. **Do not rely on MC Dropout or Deep Ensembles alone.** These methods perform worse than random on the coverage blind spot. If you use them, combine with structural coverage.
5. **For rare relations, consider flagging $c < 2$.** The coverage paradox (partial $>$ full) holds mainly for common relations. For rare relations, conventional wisdom (more coverage = better) applies.

7 Related Work

Uncertainty in KGE. UKGE [Chen et al., 2019] models confidence scores per triple, while BEUrRE [Chen et al., 2021] uses box embeddings with learned uncertainty bounds. Calibration analysis [Safavi et al., 2020] has shown KGE models are poorly calibrated. These methods focus on semantic uncertainty derived from embeddings—our work shows structural uncertainty is equally important but orthogonal, and that embedding-based methods systematically miss it.

Bayesian Deep Learning. MC Dropout [Gal and Ghahramani, 2016] approximates Bayesian inference via dropout at test time, while Deep Ensembles [Lakshminarayanan et al., 2017] capture uncertainty through model disagreement. Energy-based OOD detection [Liu et al., 2020] uses score magnitude. The epistemic-aleatoric decomposition [Kendall and Gal, 2017] distinguishes model from data uncertainty. We show these methods fail specifically on the coverage blind spot because they measure parameter uncertainty, not data coverage.

OOD Detection. Generalized OOD detection [Yang et al., 2021] encompasses various distribution shift types. Prior KG work focuses on novel entities or temporal shift. We identify a distinct failure mode—novel entity-relation *contexts*—invisible to existing embedding-based methods.

Cold-Start in KGs. Entity coverage has been studied for cold-start and inductive settings [Hamaguchi et al., 2017]. Our contribution connects coverage to uncertainty quantification and demonstrates that standard Bayesian methods are *worse than useless* (sub-random AUROC) on this failure mode—a finding not established in prior work.

8 Discussion

Why This Matters. The coverage blind spot is not an edge case. It affects 2–8% of queries and causes 67–100% error rates. More critically, standard uncertainty methods—the ones practitioners actually use—fail to detect it. Deploying KGE models with MC Dropout uncertainty could be *worse* than no uncertainty at all.

Limitations. We test on standard benchmarks; real-world industrial KGs may differ. The coverage paradox on FB15k-237 (partial > full) deserves deeper theoretical investigation. Our recommendations are empirically grounded but not formally proven optimal.

Broader Implications. The semantic-structural gap may extend beyond KGE. Any domain where models face novel *contexts* (not just novel inputs) may exhibit similar blind spots. We encourage investigation in other structured prediction settings.

9 Conclusion

We have argued that **standard uncertainty quantification methods for knowledge graph embeddings should not be trusted in isolation**. MC Dropout and Deep Ensembles capture semantic uncertainty (embedding confidence) but miss structural uncertainty (context coverage). On the coverage blind spot, these methods perform *worse than random guessing*.

The fix is simple: track entity-relation coverage with a hash table. This trivial lookup catches catastrophic failures that sophisticated Bayesian methods miss entirely.

The diagnosis matters more than the prescription. Understanding *why* standard methods fail enables practitioners to deploy KGE models responsibly. We hope this work prompts the community to reconsider the assumptions underlying uncertainty quantification in structured prediction.

Call to the Community. We invite productive disagreement: (1) Are there Bayesian methods that *do* capture structural coverage? (2) Does the coverage paradox hold in industrial KGs beyond benchmarks? (3) Can learned representations encode coverage implicitly through architectural changes? We believe debate on these questions will advance trustworthy KGE deployment.

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